

ONLINE FOOD ORDERING MANAGEMENT SYSTEM

TASK 7

AWT and JavaFX Components

- Demonstrate Different Layouts with awt and JavaFX Component.
- Design a web application with awt and JavaFX Component for event handling.

SOURCE CODE:

```
import javafx.application.Application;
```

```
import javafx.geometry.Pos;
```

```
import javafx.scene.Scene;
```

```
import javafx.scene.control.*;
```

```
import javafx.scene.layout.VBox;
```

```
import javafx.stage.Stage;
```

```
import java.awt.Frame;
```

```
import java.awt.FlowLayout;
```

```
public class CafeJavaFX extends Application
{
    Stage window; Scene slide1, slide2, slide3, slide4, slide5, slide6;
    String customerName = "";

    int total = 0;

    VBox menuBox = new VBox(10);

    Label billLabel = new Label("Total Bill: ₹0");

    /* Veg Menu */
    CheckBox veg1 = new CheckBox("Veg Biryani ₹120");
    TextField vegQty1 = new TextField("0");
    CheckBox veg2 = new CheckBox("Veg Rice ₹90"); TextField
    vegQty2 = new TextField("0");
    CheckBox veg3 = new CheckBox("Paneer Pulao ₹130");
    TextField vegQty3 = new TextField("0");
    CheckBox veg4 = new CheckBox("Veg Noodles ₹100");
    TextField vegQty4 = new TextField("0");
    CheckBox veg5 = new CheckBox("Veg Sandwich ₹80");
    TextField vegQty5 = new TextField("0");
```

```
/* Non Veg */
```

```
CheckBox non1 = new CheckBox("Chicken Biryani ₹180");
```

```
TextField nonQty1 = new TextField("0");
```

```
CheckBox non2 = new CheckBox("Mutton Biryani ₹220");
```

```
TextField nonQty2 = new TextField("0");
```

```
CheckBox non3 = new CheckBox("Chicken Noodles ₹150");
```

```
TextField nonQty3 = new TextField("0");
```

```
CheckBox non4 = new CheckBox("Chicken Rice ₹140");
```

```
TextField nonQty4 = new TextField("0");
```

```
CheckBox non5 = new CheckBox("Non Veg Meals ₹200");
```

```
TextField nonQty5 = new TextField("0");
```

```
/* Beverages */
```

```
CheckBox bev1 = new CheckBox("Cool Drink ₹40");
```

```
TextField bevQty1 = new TextField("0");
```

```
CheckBox bev2 = new CheckBox("Milkshake ₹70");
```

```
TextField bevQty2 = new TextField("0");
```

```
CheckBox bev3 = new CheckBox("Fruit Juice ₹60");
```

```
TextField bevQty3 = new TextField("0");
CheckBox bev4 = new CheckBox("Lime Juice ₹30");

TextField bevQty4 = new TextField("0");
CheckBox bev5 = new CheckBox("Faluda ₹90");

TextField bevQty5 = new TextField("0");

/* AWT Welcome Window */
public static void showAWT(String[] args){
    Frame frame = new Frame("Cafe Food Ordering System");
    frame.setSize(300,200);
    frame.setLayout(new FlowLayout());
    java.awt.Label label = new java.awt.Label("Welcome to Cafe
Food Ordering System");
    java.awt.Button start = new java.awt.Button("Start");
    start.addActionListener(e -> {
        frame.dispose();
        launch(args);
    });
    frame.add(label);
```

```
        frame.add(start);
        frame.setVisible(true);
    }

    @Override
    public void start(Stage stage){
        window = stage;
        addAutoListeners();

        /* Slide 1 */
        Label welcomeLabel = new Label("Welcome to
Online Food Ordering");
        Button nextBtn1 = new Button("Next");
        VBox layout1 = new VBox(20);
        layout1.getChildren().addAll(welcomeLabel,
nextBtn1);
        layout1.setAlignment(Pos.CENTER);
        slide1 = new Scene(layout1, 450, 300);

        /* Slide 2 */
        Label nameLabel = new Label("Enter Your
```

```
Name");
    TextField nameField = new TextField();
    Button nextBtn2 = new Button("Next");
    VBox layout2 = new VBox(20);
    layout2.getChildren().addAll(nameLabel,
nameField, nextBtn2);
    layout2.setAlignment(Pos.CENTER);
    slide2 = new Scene(layout2, 450, 300);

    /* Slide 3 */
    CheckBox vegBox = new CheckBox("Veg");
    CheckBox nonVegBox = new CheckBox("Non
Veg");
    CheckBox bevBox = new
CheckBox("Beverages");
    Button prevBtn3 = new Button("Previous");
    Button nextBtn3 = new Button("Next");
    VBox layout3 = new VBox(15);
    layout3.getChildren().addAll(new
Label("Select Category"), vegBox, nonVegBox,
bevBox, prevBtn3, nextBtn3);
```

```
layout3.setAlignment(Pos.CENTER);
slide3 = new Scene(layout3, 450, 350);

/* Slide 4 */
Button prevBtn4 = new Button("Previous");
Button nextBtn4 = new Button("Next");
menuBox.setAlignment(Pos.CENTER);
VBox layout4 = new VBox(10);
layout4.getChildren().addAll(new
Label("Menu"), menuBox, billLabel, prevBtn4,
nextBtn4);
layout4.setAlignment(Pos.CENTER);
slide4 = new Scene(layout4, 450, 500);

/* Slide 5 */
Button prevBtn5 = new Button("Previous");
Button nextBtn5 = new Button("Next");
VBox layout5 = new VBox(20);
layout5.getChildren().addAll(new
Label("Bill Summary"), billLabel, prevBtn5,
nextBtn5);
```

```
layout5.setAlignment(Pos.CENTER);
slide5 = new Scene(layout5, 450, 300);

/* Slide 6 */
Label thankLabel = new Label("Thank You!
Visit Again");
Button prevBtn6 = new Button("Previous");
VBox layout6 = new VBox(20);
layout6.getChildren().addAll(thankLabel,
prevBtn6);
layout6.setAlignment(Pos.CENTER);
slide6 = new Scene(layout6, 450, 300);

/* Actions */
nextBtn1.setOnAction(e ->
window.setScene(slide2));
nextBtn2.setOnAction(e -> { customerName =
nameField.getText();
window.setScene(slide3); });
prevBtn3.setOnAction(e ->
window.setScene(slide2));
```



```
        nextBtn3.setOnAction(e -> {
            menuBox.getChildren().clear();
            if(vegBox.isSelected())
menuBox.getChildren().addAll(veg1, vegQty1,
veg2, vegQty2, veg3, vegQty3, veg4, vegQty4,
veg5, vegQty5);
            if(nonVegBox.isSelected())
menuBox.getChildren().addAll(non1, nonQty1,
non2, nonQty2, non3, nonQty3, non4, nonQty4,
non5, nonQty5);
            if(bevBox.isSelected())
menuBox.getChildren().addAll(bev1, bevQty1,
bev2, bevQty2, bev3, bevQty3, bev4, bevQty4,
bev5, bevQty5);
            updateBill();
            window.setScene(slide4);
        });
        prevBtn4.setOnAction(e ->
window.setScene(slide3));
        nextBtn4.setOnAction(e -> { updateBill();
```

```
window.setScene(slide5); });  
    prevBtn5.setOnAction(e ->  
window.setScene(slide4));  
    nextBtn5.setOnAction(e ->  
window.setScene(slide6));  
    prevBtn6.setOnAction(e ->  
window.setScene(slide5));  
  
    window.setTitle("Cafe Food Order System");  
    window.setScene(slide1);  
    window.show();  
}
```

```
private void addAutoListeners(){  
    CheckBox[] boxes = {veg1, veg2, veg3, veg4, veg5,  
non1, non2, non3, non4, non5, bev1, bev2, bev3, bev4, bev5};  
    TextField[] qty = {vegQty1, vegQty2, vegQty3, vegQty4,  
vegQty5, nonQty1, nonQty2, nonQty3, nonQty4, nonQty5,  
bevQty1, bevQty2, bevQty3, bevQty4, bevQty5};  
    for(CheckBox cb : boxes)
```

```
cb.setAction(e -> updateBill());
```

```
for(TextField tf : qty)
```

```
tf.textProperty().addListener((a,b,c) -> updateBill());
```

```
}
```

```
private int safeParse(TextField tf){ try{ return
```

```
Integer.parseInt(tf.getText()); } catch(Exception e){ return 0; } }
```

```
private void updateBill(){
```

```
    total=0;
```

```
    if(veg1.isSelected()) total+=safeParse(vegQty1)*120;
```

```
    if(veg2.isSelected()) total+=safeParse(vegQty2)*90;
```

```
    if(veg3.isSelected()) total+=safeParse(vegQty3)*130;
```

```
    if(veg4.isSelected()) total+=safeParse(vegQty4)*100;
```

```
    if(veg5.isSelected()) total+=safeParse(vegQty5)*80;
```

```
    if(non1.isSelected()) total+=safeParse(nonQty1)*180;
```

```
    if(non2.isSelected()) total+=safeParse(nonQty2)*220;
```

```
    if(non3.isSelected()) total+=safeParse(nonQty3)*150;
```

```
    if(non4.isSelected()) total+=safeParse(nonQty4)*140;
```

```
    if(non5.isSelected()) total+=safeParse(nonQty5)*200;
```

```
    if(bev1.isSelected()) total+=safeParse(bevQty1)*40;
```

```

        if(bev2.isSelected()) total+=safeParse(bevQty2)*70;
        if(bev3.isSelected()) total+=safeParse(bevQty3)*60;
        if(bev4.isSelected()) total+=safeParse(bevQty4)*30;
        if(bev5.isSelected()) total+=safeParse(bevQty5)*90;
        billLabel.setText("Customer: "+customerName+"\nTotal Bill:
₹"+total);
    }

    public static void main(String[] args)
    {
        showAWT(args);
    }
}

```

TERMINAL SYNTAX:

```

PS C:\VTP4906\online food order> javac --module-path javafx-
sdk-21.0.11/lib --add-modules javafx.controls CafeJavaFX.java

```

```

PS C:\VTP4906\online food order> java --module-path javafx-
sdk-21.0.11/lib --add-modules javafx.controls CafeJavaFX

```

OUTPUT :

Cafe Food Ordering System

Welcome to Cafe Food Ordering System

Start

Cafe Food Order System

Enter Your Name

PRIYANKA

Next

Cafe Food Order System

Select Category

☒ Veg

☐ Non Veg

☐ Beverages

Previous

Next

Cafe Food Order System

Menu

☐ Veg Biryani ₹120

☐ Veg Rice ₹90

☐ Paneer Pulao ₹130

☒ Veg Noodles ₹100

☐ Veg Sandwich ₹80

0

0

0

2

0

Previous

Next

Cafe Food Order System

Bill Summary

Customer: PRIYANKA

Total Bill: ₹200

Previous

Next

Cafe Food Order System

Thank You! Visit Again

Previous